

01_HCAT_Taxonomy

HCAT Taxonomy Documentation

HCAT Insight — Rassoul Azam Hospital

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1. Introduction

This document provides a complete, authoritative reference for the classification taxonomy used within **HCAT Insight**, the healthcare complaint management and AI-assisted analysis platform deployed at Rassoul Azam Hospital (RAH).

The taxonomy defines how patient complaints are structured, labelled, and interpreted across nine classification dimensions. It serves as the semantic foundation of the entire HCAT Insight system: every workflow routing decision, every AI model prediction, and every analytical report is anchored to this taxonomy.

This document covers:

- The origin and purpose of the classification framework
- The definition and operational meaning of every class in every dimension
- The hierarchical dependency structure between dimensions
- The actual data distributions observed in the RAH complaint dataset
- Challenges in annotation and label consistency
- How the taxonomy is used operationally and by AI/ML components

2. HCAT Framework Background

The **Healthcare Complaints Analysis Tool (HCAT)** is an internationally recognized framework for the systematic classification and analysis of patient complaints within healthcare institutions. It was developed to provide a standardized vocabulary for capturing the nature, severity, and context of complaints, enabling healthcare institutions to identify patterns, benchmark performance, and drive quality improvement.

The HCAT framework organizes complaints across a set of structured dimensions:

- A broad **Domain** that indicates the general area of care affected
- A **Category** that specifies the type of failure within that domain
- More granular **Sub-Category** and fine-grained classification labels
- Severity and harm assessments that capture the clinical impact of the incident
- Stage of care labels that identify where in the care pathway the issue arose

HCAT has been applied in multiple international studies. A relevant 2025 study published in JMIR (Koh et al.) demonstrated that large language models including GPT-4o and Claude 3.5 Sonnet could classify patient complaints according to HCAT with a mean concordance of approximately 68.8% in zero-shot conditions, confirming both the viability of automated classification and the inherent complexity of the taxonomy.

3. RAH Adaptation Context

Rassoul Azam Hospital (RAH) is a specialized cardiac care hospital in which the HCAT taxonomy has been adapted and operationalized in Arabic for local complaint processing.

Key adaptation characteristics:

- Language:** All complaint text is written in Arabic, with occasional mixed Arabic-English clinical terminology
- Dataset:** 476 cleaned complaint records collected starting from January 2025
- Extended taxonomy:** RAH extends the base HCAT dimensions with a fine-grained Arabic/English classification layer of 78 Arabic labels and 73 English labels
- Operational fields:** RAH adds operational fields not present in base HCAT, including Improvement Opportunity Type (Red Flag, Never Event) and Complaint Status

The taxonomy as implemented in HCAT Insight is used simultaneously for:

- Operational routing and escalation of complaints within the hospital

2. Training and evaluation of AI/ML classification models
3. Analytical reporting and trend identification

4. Taxonomy Architecture Overview

The HCAT taxonomy as implemented in HCAT Insight contains nine classification dimensions. Three of these dimensions form a strict hierarchical tree (Domain → Category → Sub-Category). The remaining dimensions are semi-independent assessments that carry contextual dependencies on the primary three.

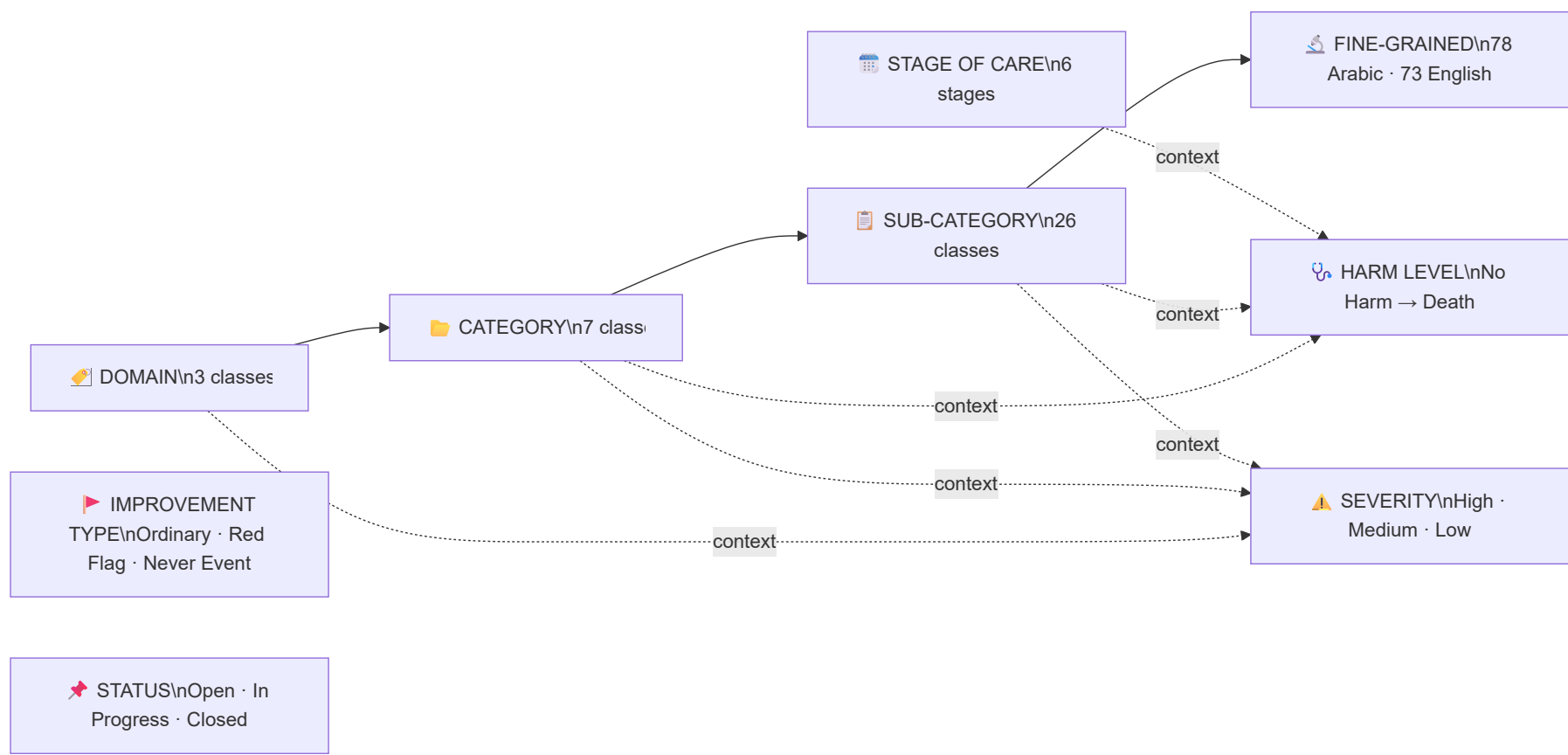


Figure 1: HCAT Insight Taxonomy Architecture. Solid arrows = strict hierarchical containment. Dashed arrows = contextual dependency used by AI/ML models.

5. Primary Classification Dimensions

The three primary dimensions form the semantic core of the HCAT taxonomy. They are organized as a strict tree: each Domain contains specific Categories, and each Category contains specific Sub-Categories. This structure is non-overlapping — a complaint belongs to exactly one path through the tree.

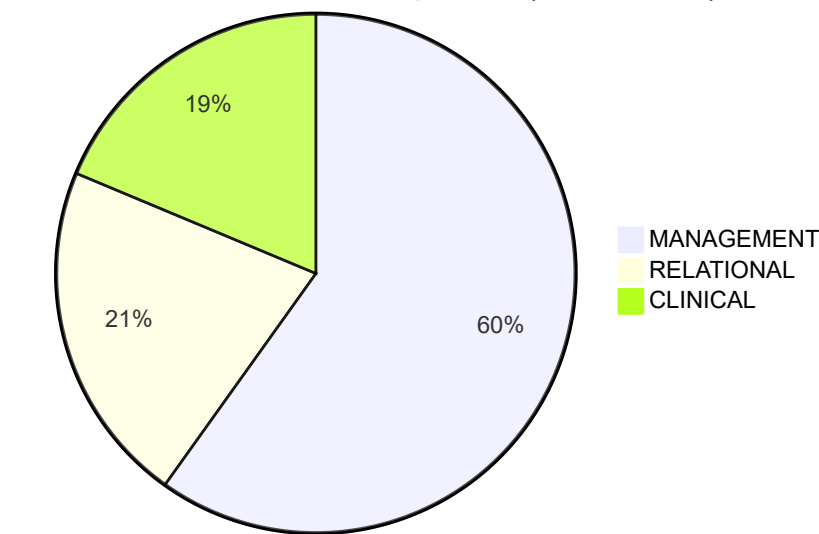
5.1 Domain

The Domain is the top-level classification that identifies the broad area of healthcare activity in which the complaint originates.

Domain	Definition
CLINICAL	Complaints arising from the clinical care process itself: diagnosis, treatment planning, nursing procedures, medications, medical decisions, and any direct patient care activity
MANAGEMENT	Complaints arising from organizational systems, administrative processes, the physical environment, resource availability, and institutional workflows
RELATIONAL	Complaints arising from the interpersonal behavior of staff toward patients and their families: communication failures, unwillingness to listen, and violations of dignity or patient rights

Distribution in RAH Dataset (n = 476):

in Distribution — 476 Complaints (RAH, 2025)



Domain	Count	Percentage
MANAGEMENT	285	59.9%
RELATIONAL	102	21.4%
CLINICAL	89	18.7%

Operational Significance: MANAGEMENT is the dominant domain, reflecting that RAH complaints predominantly concern organizational and process issues rather than direct clinical care failures. CLINICAL complaints, while least frequent, carry the highest clinical risk and require prioritized investigation.

5.2 Category

The Category refines the Domain classification into a specific type of failure. Each Domain maps to a fixed, non-overlapping set of Categories.

Domain-to-Category Mapping:

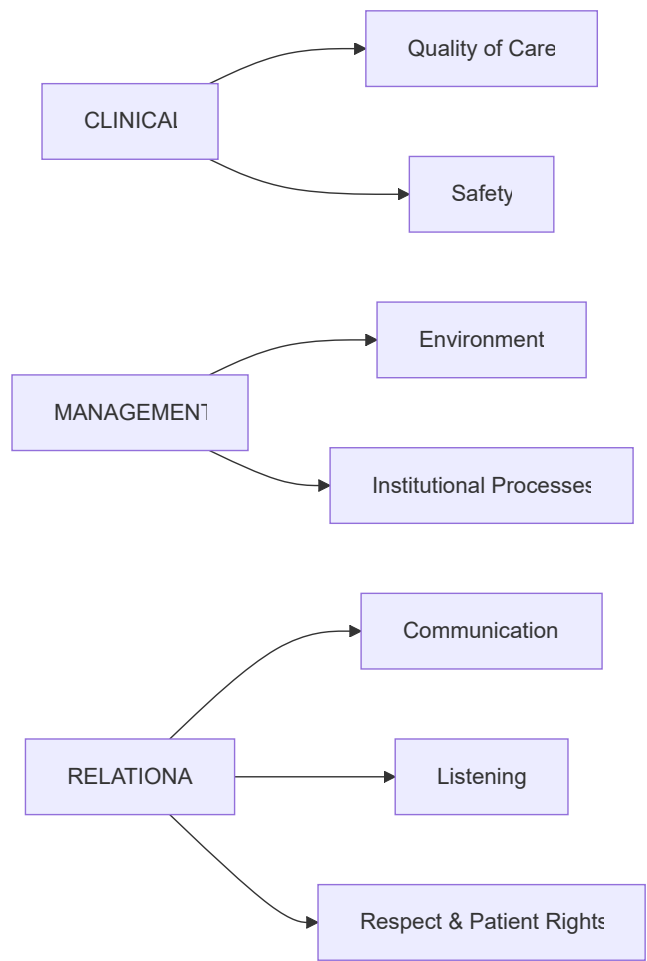
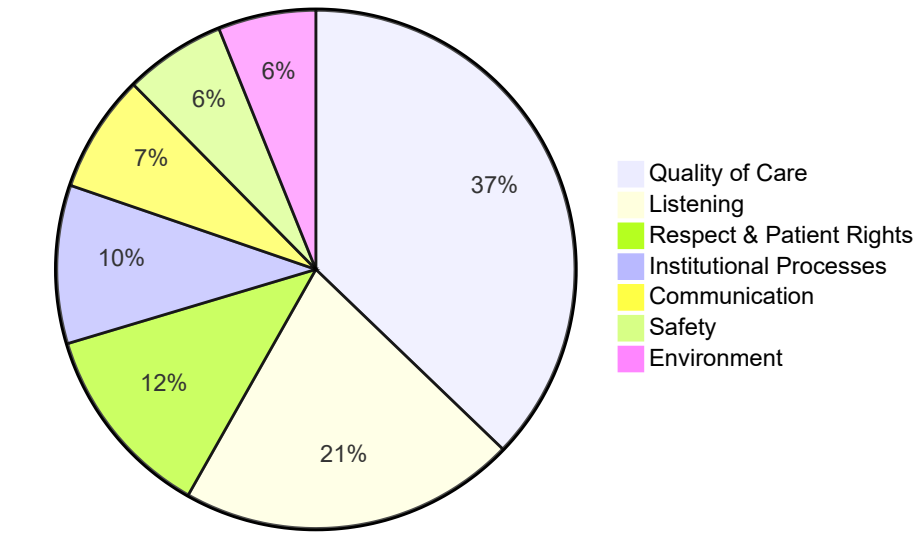


Figure 2: Domain-to-Category Hierarchy.

Category	Parent Domain	Definition
Quality of Care	CLINICAL	Issues with the standard or quality of medical and nursing care delivered, including examination, monitoring, and basic care procedures
Safety	CLINICAL	Events where patient safety was directly compromised or threatened, including clinical errors, medication errors, and teamwork failures
Environment	MANAGEMENT	Issues with the physical environment, accommodation conditions, equipment availability, and cleanliness of facilities
Institutional Processes	MANAGEMENT	Issues with organizational workflows, administrative procedures, access delays, documentation, and bureaucratic obstacles
Communication	RELATIONAL	Failures in information exchange between staff and patients or families, including absent, delayed, or incorrect communication
Listening	RELATIONAL	Situations where staff failed to acknowledge, hear, or meaningfully respond to patient concerns or complaints
Respect & Patient Rights	RELATIONAL	Incidents where patient dignity, personal beliefs, or institutional rights were not upheld

Distribution in RAH Dataset:

Category Distribution — 476 Complaints (RAH, 2025)



Category	Count	Percentage
Quality of Care	177	37.2%
Listening	100	21.0%
Respect & Patient Rights	58	12.2%
Institutional Processes	47	9.9%
Communication	35	7.4%
Safety	30	6.3%
Environment	29	6.1%

Operational Significance: Quality of Care dominates, followed by Listening — together accounting for nearly 60% of all complaints. Safety complaints, while the least frequent, carry the highest clinical significance and require immediate review regardless of volume.

5.3 Sub-Category

The Sub-Category provides granular identification of the specific nature of the complaint within its Category. Sub-Categories are strictly contained within their parent Category and do not overlap across Categories.

Complete Sub-Category Hierarchy:

Figure 3 — Full three-level taxonomy tree (Domain → Category → Sub-Category).
Generated externally. Use the prompt below to reproduce this figure.

GPT Image Generation Prompt for Figure 3:

Create a professional, publication-quality hierarchical tree diagram for a healthcare complaint classification system called HCAT Insight. The diagram has three levels: Domain, Category, and Sub-Category. Use a clean white background with a minimal design suitable for an academic paper. Use three distinct soft colors — one per Domain — that flow through the Category and Sub-Category nodes to show group membership. Use rounded rectangles for nodes. Font should be clean and sans-serif. Arrows should be thin and directional (parent to child).

The hierarchy is:

DOMAIN: CLINICAL (blue tones)

- Category: Quality of Care → Sub-Categories: Neglect — General, Examination & Monitoring, Neglect — Hygiene & Personal Care
- Category: Safety → Sub-Categories: Clinician Errors, Error — Diagnosis, Error — General, Error — Medication, Failure to Respond, Teamwork

DOMAIN: MANAGEMENT (green tones)

- Category: Environment → Sub-Categories: Accommodation, Equipment, Ward Cleanliness
- Category: Institutional Processes → Sub-Categories: Bureaucracy, Delay — Access, Delay — General, Delay — Procedure, Documentation, Visiting

DOMAIN: RELATIONAL (orange tones)

- Category: Communication → Sub-Categories: Absent Communication, Delayed Communication, Failure to Provide, Incorrect Communication
- Category: Listening → Sub-Categories: Dismissing Patients, Ignoring Patients
- Category: Respect & Patient Rights → Sub-Categories: Disrespect, Rights

Layout: horizontal tree expanding left to right. Domain nodes on the left, Category nodes in the middle column, Sub-Category nodes on the right. Title at the top: "HCAT Insight — Three-Level Complaint Taxonomy". Include a small legend indicating the three Domain color groups. Style: clean, academic, suitable for a healthcare informatics journal.

Figure 3: Complete Sub-Category Hierarchy by Domain and Category.

Sub-Category Definitions:

Sub-Category	Parent Category	Definition
Neglect — General	Quality of Care	General failure to provide expected level of nursing or clinical care
Examination & Monitoring	Quality of Care	Inadequate, incorrect, or absent patient examination or clinical monitoring
Neglect — Hygiene & Personal Care	Quality of Care	Failure to maintain patient hygiene, personal care, or basic comfort needs
Clinician Errors	Safety	Clinical mistakes made by physicians or specialists that compromise care quality
Error — Diagnosis	Safety	Incorrect, missed, or delayed diagnosis
Error — General	Safety	General clinical errors not specifically categorized under other safety sub-types
Error — Medication	Safety	Incorrect medication, dose, timing, or administration errors
Failure to Respond	Safety	Failure of nursing or clinical staff to respond to patient calls or emergencies
Teamwork	Safety	Coordination failures between clinical team members that affected patient care
Accommodation	Environment	Issues with patient room conditions, bed availability, privacy, ventilation, or comfort
Equipment	Environment	Missing, malfunctioning, or insufficient medical or non-medical equipment
Ward Cleanliness	Environment	Complaints about the hygiene and cleanliness of rooms, bathrooms, or common areas
Bureaucracy	Institutional Processes	Excessive administrative procedures, approvals, or obstacles blocking timely care
Delay — Access	Institutional Processes	Delays in accessing care, appointments, or consultations
Delay — General	Institutional Processes	General delays in hospital services not fitting a more specific sub-type
Delay — Procedure	Institutional Processes	Delays in the execution of a specific medical or nursing procedure
Documentation	Institutional Processes	Issues with medical records, discharge documentation, or administrative file preparation
Visiting	Institutional Processes	Complaints about visiting policies, visiting hours, or visitor restrictions
Absent Communication	Communication	Complete absence of communication from staff regarding the patient's condition or plan
Delayed Communication	Communication	Communication that occurred too late to be useful or to prevent distress
Failure to Provide	Communication	Staff failing to provide required information, instructions, or explanations
Incorrect Communication	Communication	Staff providing inaccurate, misleading, or conflicting information
Dismissing Patients	Listening	Staff visibly dismissing or minimizing patient concerns without engaging with them
Ignoring Patients	Listening	Staff failing to acknowledge patient presence, calls, or requests
Disrespect	Respect & Patient Rights	Rude, demeaning, or unprofessional behavior toward patients or their families
Rights	Respect & Patient Rights	Violations of patient rights including religious beliefs, consent, or institutional rights

Total: 26 Sub-Categories across 7 Categories.

6. Fine-Grained Classification Layer

Beyond the three-level primary hierarchy, HCAT Insight includes an extended fine-grained classification layer that further specifies the exact nature of the complaint. This layer was developed by RAH to support more detailed operational analysis and routing.

- Arabic Classification Labels: 78 distinct classes in Arabic
- English Classification Labels: 73 distinct classes in English

This layer captures highly specific complaint types such as:

- IV insertion errors, imaging appointment delays, discharge paperwork problems
- Specific noise complaints (equipment noise vs. workshop noise vs. staff noise)
- Specific accommodation sub-types (air conditioning, bed availability, hygiene supplies)

Operational Role: The fine-grained layer is used for detailed internal reporting and root-cause analysis. Due to the large number of classes and the limited dataset size (476 records), this layer is not currently targeted by AI classification models.

7. Complaint Severity Level

The Severity Level is an ordinal assessment of the intensity, impact, and urgency of the complaint. It is assigned by the complaint officer at the time of case intake, based on the complaint narrative and supporting context.

Level	Definition	Operational Response
HIGH	The complaint describes a serious incident with significant impact on patient care, safety, or institutional reputation. Requires urgent attention.	Immediate escalation; senior management notification
MEDIUM	The complaint describes a significant issue that caused notable patient distress or disruption to care. Requires structured follow-up.	Priority review within defined timeframe
LOW	The complaint describes a less serious issue, minor inconvenience, or service dissatisfaction without direct care impact.	Standard resolution workflow

Distribution in RAH Dataset:

Level	Count	Percentage
LOW	317	66.6%
MEDIUM	125	26.3%
HIGH	28	5.9%

Note: A small number of records (n=6) carry a legacy severity label ("Moderate") that predates the standardized three-level scale. These were retained in the cleaned dataset for model training purposes and treated as a MEDIUM equivalent.

Clinical Significance: Severity assessment is inherently subjective and context-dependent. It is influenced by the Domain, Category, and Sub-Category of the complaint. A Delay in access for a routine appointment (LOW) and a Delay in access during an acute emergency (HIGH) share the same Sub-Category but differ dramatically in severity. This context-dependence is one of the primary reasons Severity is among the most challenging features to classify automatically.

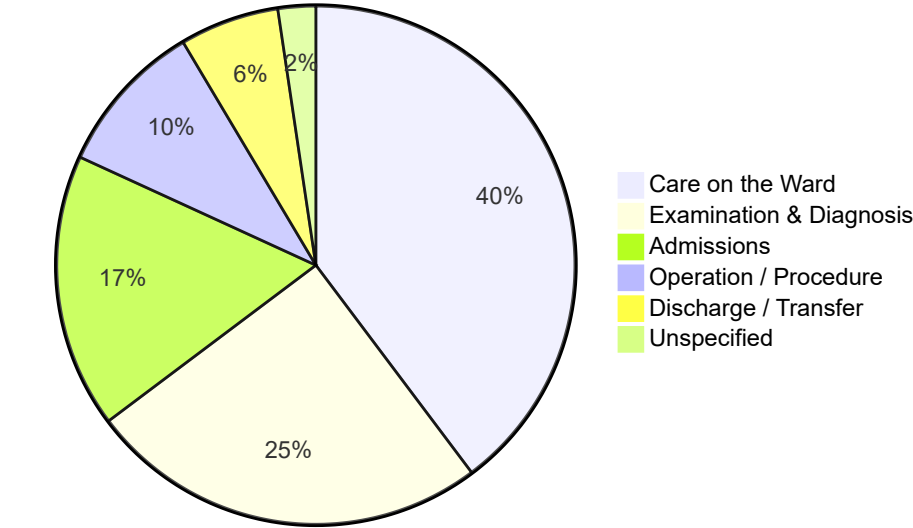
8. Stage of Care

The Stage of Care identifies the point in the patient care pathway at which the complaint-generating event occurred. This information is used to identify systemic failures at specific care stages and to route complaints to the appropriate clinical or administrative team.

Stage	Definition
Admissions	Issues occurring during the patient registration, intake, and admission process
Examination & Diagnosis	Issues occurring during clinical examination, diagnostic workup, or test result communication
Care on the Ward	Issues occurring during inpatient care on the ward, including nursing care, daily rounds, and in-room services
Operation / Procedure	Issues occurring during a surgical operation or medical intervention procedure
Discharge / Transfer	Issues occurring at the time of patient discharge or transfer to another unit or facility
Unspecified	The stage cannot be determined from the available complaint narrative

Distribution in RAH Dataset:

Stage of Care Distribution — 468 Records (RAH, 2025)



Stage	Count	Percentage
Care on the Ward	186	39.1%
Examination & Diagnosis	117	24.6%
Admissions	80	16.8%
Operation / Procedure	45	9.5%
Discharge / Transfer	29	6.1%
Unspecified	11	2.3%

Note: Care on the Ward is the dominant stage, reflecting the extended duration of inpatient experience and the multiple opportunities for service failures during hospitalization.

AI Classification Significance: Stage classification presents a unique challenge because a single complaint may reference events across multiple stages. A complaint about receiving incorrect discharge instructions, for example, may also reference care quality issues that began on the ward. The HCAT Insight AI pipeline uses a dedicated semantic projection approach for Stage prediction rather than a standard classification model, due to the multi-stage nature of many complaints.

9. Harm Level

The Harm Level assesses the actual clinical impact of the complaint event on the patient. Unlike Severity (which reflects perceived urgency), Harm reflects the objective clinical consequence — the damage, if any, that occurred.

Level	Definition
No Harm	The incident occurred or the complaint concern was raised, but no physical or psychological harm resulted to the patient
Minor Harm	The patient experienced temporary, minor harm or distress that required minimal or no additional clinical intervention

Level	Definition
Moderate Harm	The patient experienced moderate harm requiring significant additional intervention or causing meaningful distress or delay in recovery
Severe Harm	The patient experienced permanent or life-altering harm as a result of the incident
Death	The incident resulted in or contributed to the death of the patient

Distribution in RAH Dataset:

Level	Count	Percentage
No Harm	122	25.6%
Minor Harm	28	5.9%
Moderate Harm	7	1.5%
Severe Harm	186	39.1%
Death	128	26.9%

Important Contextual Note: The distribution of Harm Level in the RAH dataset reflects the operational context of HCAT Insight as deployed within a specialized cardiac care hospital. Cardiac patients present inherently with serious underlying conditions, and the hospital manages a high acuity patient population. The presence of complaints involving Severe Harm and Death categories is expected in this clinical environment and reflects the severity of the patients' baseline conditions, not necessarily institutional failures.

AI Classification Significance: Harm is considered the most challenging classification target in the HCAT framework. Human annotators have historically shown lower inter-rater agreement on Harm than on any other dimension, because determining harm requires knowledge of:

- What actually happened to the patient (not just what they reported)
- The patient's baseline clinical condition
- The counterfactual — what would have happened without the incident

The AI system addresses this by using both patient complaint text and hospital administration response text as inputs, and by implementing a two-stage architecture: a binary High-Harm versus Low-Harm classifier for safety triage, combined with a fine-grained 5-class classifier for reporting purposes.

10. Improvement Opportunity Type

This dimension classifies the operational priority and escalation level of the complaint, distinct from its clinical content.

Type	Definition	Frequency
Ordinary Complaint	A standard complaint that follows the normal complaint resolution workflow	92.6% (n=441)
Red Flag	A complaint requiring urgent escalation to hospital management, outside the normal resolution workflow. Applied when the complaint content or context demands immediate management attention.	4.0% (n=19)
Never Event	A complaint involving a serious adverse event that should never occur in a properly functioning healthcare setting, by international standards. Triggers a formal investigation protocol.	< 1% (from dataset header; encoded separately)

Operational Role: This dimension directly governs workflow routing in HCAT Insight. Red Flag complaints trigger immediate notification to designated management personnel. Never Events activate a formal root-cause analysis process. Ordinary Complaints enter the standard queue.

AI Classification Significance: The extreme class imbalance (92.6% Ordinary Complaint) makes this a difficult classification target despite high overall accuracy. The AI model achieves ~97.8% accuracy but this is largely driven by the dominant class. Red Flag and Never Event detection from text alone remains a research challenge.

11. Complaint Status

Status tracks the lifecycle position of a complaint within the HCAT Insight workflow.

Status	Definition
Open	Complaint received and entered into the system but not yet assigned for investigation
In Progress	Complaint under active investigation, resolution, or follow-up
Closed	Complaint fully resolved, response provided, and case documented

Status is an operational field managed by users in the HCAT Insight workflow system. It does not function as an AI classification target.

12. Label Dependency Structure

One of the most important discoveries in the HCAT Insight AI research is that the nine classification dimensions are **not independent**. They form a structured dependency graph that mirrors the natural cognitive process a human annotator follows when classifying a complaint.

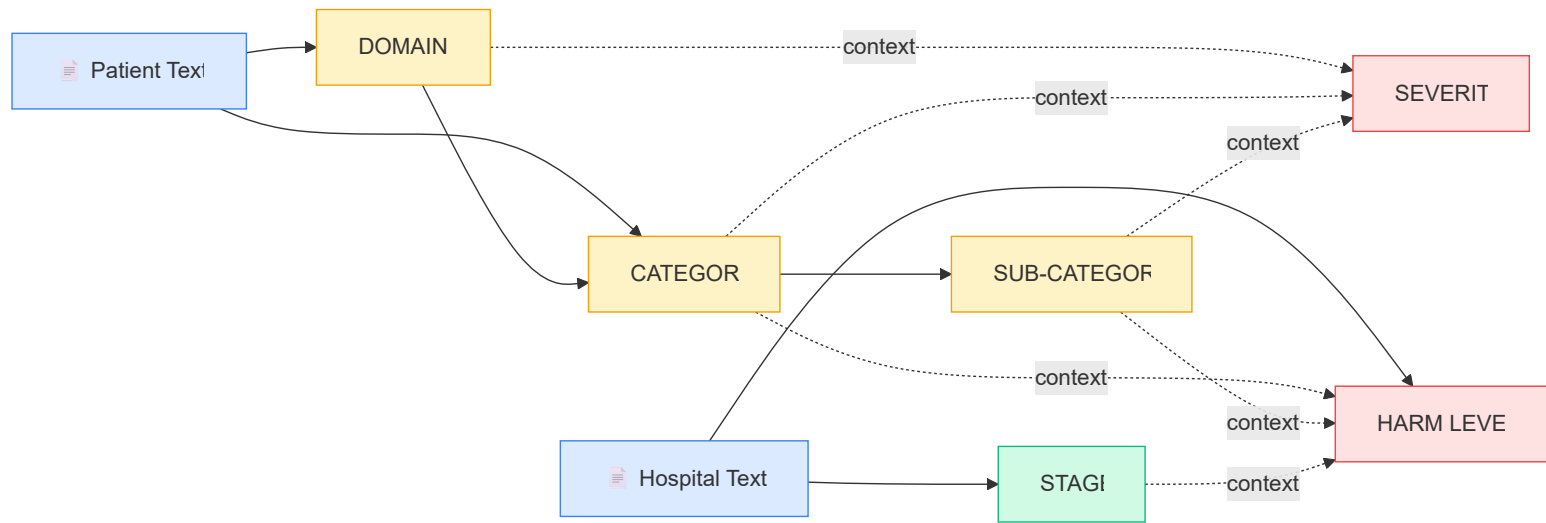


Figure 4: Label Dependency Graph for HCAT Insight AI Pipeline.

Key Structural Properties:

Property	Description
Domain is independent	Domain is predicted solely from patient text with no dependence on other labels
Category depends on Domain	Each Domain admits only specific Categories; Category predictions conditioned on Domain are more accurate
Sub-Category depends on Category	Sub-Category is non-overlapping across Categories; conditioning reduces the classification space dramatically
Severity is derived	Severity depends on Domain, Category, Sub-Category, and text features simultaneously
Stage is semi-independent	Stage correlates with Domain and Category but primarily signals from hospital administration text
Harm is the most complex	Harm depends on all upstream labels plus both patient and hospital text; it is the terminal node of the dependency graph

Annotation Order Insight: Human annotators at RAH implicitly follow this dependency order when classifying complaints:

1. What area of care does this concern? → Domain
2. What type of failure is this? → Category
3. What specifically happened? → Sub-Category
4. How bad was it? → Severity
5. Where in the care pathway? → Stage
6. What actually happened to the patient? → Harm

This cognitive ordering is the basis for the HCAT Insight hierarchical AI architecture.

13. Data Distribution Summary

Dataset Overview:

Property	Value
Total Records (Clean)	476
Training Set	380 (79.8%)
Test Set	96 (20.2%)
Language	Arabic (with mixed English clinical terms)
Date Range	January 2025 onwards
Hospital	Rassoul Azam Hospital (RAH) — Specialized Cardiac Care

Classification Dimension Summary:

Dimension	Classes	Most Frequent Class	Dominant %
Domain	3	MANAGEMENT	59.9%
Category	7	Quality of Care	37.2%
Sub-Category	26	Documentation	—
Severity	3	LOW	66.6%
Stage	6	Care on the Ward	39.1%
Harm	5	Severe Harm	39.1%
Improvement Type	3	Ordinary Complaint	92.6%

Class Imbalance: All dimensions exhibit significant class imbalance. This is an inherent property of real operational complaint data and has direct implications for AI model training, evaluation, and deployment. The macro-averaged F1 score is the primary metric used for all models to counteract the effect of dominant classes.

14. Annotation Challenges and Data Quality

14.1 Label Ambiguity

Several sub-categories overlap semantically, creating annotation ambiguity. The most notable overlaps are:

- **Communication vs. Listening:** Both concern staff-patient interaction. Communication focuses on information exchange failures; Listening focuses on the staff's willingness to engage. However, in practice, a complaint about a doctor not explaining the treatment plan can be interpreted as either.
- **Clinician Errors vs. Error — General:** The boundary between a specific clinician's mistake and a general clinical error is not always clear, particularly when the complaint narrative is brief.
- **Delay — General vs. Delay — Procedure vs. Delay — Access:** The three delay sub-categories are semantically close. Without temporal context clues in the text, distinguishing them requires inference.

14.2 Multi-Stage Complaints

Many complaints reference events that occurred across multiple stages of care. Because the annotation schema forces a single-label assignment for Stage, annotators must select the most clinically significant stage. This introduces noise into stage labels and is a recognized limitation of the HCAT protocol as applied to narrative complaints.

14.3 Data Quality Issues Encountered and Resolved

During preprocessing, the following data quality issues were identified and resolved:

Issue	Description	Resolution
Inconsistent severity casing	Values such as "MEDIUM", "Medium", and "mEDIUM" present simultaneously	Standardized to three canonical values: HIGH, MEDIUM, LOW
Trailing whitespace in stage labels	"Admissions " and "Admissions" treated as different classes	Stripped during preprocessing
Legacy harm label	"High Severe" present as an additional harm level	Mapped to Severe Harm during cleaning
Duplicate sub-category names	Case variants such as "Delay -Access" and "Delay -access"	Unified to canonical form
Null stage values	8 records with missing Stage label	Retained in dataset; excluded from Stage model training

15. Operational Usage

The HCAT taxonomy governs four operational functions within HCAT Insight:

Function	Dimensions Used
Complaint Routing	Domain, Category, Sub-Category → determines which team receives the case
Priority Escalation	Improvement Opportunity Type, Severity → determines urgency and escalation path
Safety Monitoring	Harm Level, Severity, Never Event → triggers safety review processes
Reporting & Analytics	All dimensions → aggregated in monthly and seasonal reports for management

The taxonomy also defines the structure of all dashboard metrics, trend charts, and comparative analyses produced by the HCAT Insight reporting system.

16. AI/ML Integration

The HCAT taxonomy directly shapes the architecture of the HCAT Insight AI classification system.

AI Component	Taxonomy Dimension	Architecture Notes
Domain Classifier	Domain (3 classes)	Single-head Logistic Regression on patient text embeddings; best-performing model
Category Classifier	Category (7 classes)	Hierarchical model conditioned on predicted Domain; separate classifier per Domain
Sub-Category Classifier	Sub-Category (26 classes)	Hierarchical model conditioned on predicted Category; separate classifier per Category
Severity Classifier	Severity (3 levels)	Ordinal Logistic Regression using combined patient + hospital text embeddings
Stage Classifier	Stage (6 stages)	Semantic projection model using vocab-based metric embeddings; hospital text primary
Harm Classifier	Harm (5 levels)	Two-stage: binary safety triage + fine-grained 5-class model; all embeddings as input
Improvement Type	IOT (3 types)	Single-head classifier; dominated by Ordinary Complaint class

All text inputs are encoded using the `sentence-transformers/paraphrase-multilingual-mpnet-base-v2` model, which produces 768-dimensional dense embeddings capable of representing Arabic clinical text effectively without fine-tuning.

The dependency structure described in Section 12 directly informs the hierarchical and stacked model architecture — where predictions from upstream models (Domain, Category) are fed as additional input features to downstream models (Severity, Harm).

17. Conclusion

The HCAT taxonomy as implemented in HCAT Insight is a multi-dimensional, hierarchically structured classification system adapted from the international HCAT framework for the Arabic-language operational environment of Rassoul Azam Hospital.

Key characteristics of this taxonomy:

- Three-level primary hierarchy:** Domain → Category → Sub-Category, forming a strict non-overlapping tree with $3 \times 7 \times 26$ nodes
- Extended fine-grained layer:** 78 Arabic and 73 English labels for detailed operational analysis
- Four contextual assessment dimensions:** Severity, Stage, Harm, and Improvement Opportunity Type
- Non-independence of labels:** Labels form a structured dependency graph that mirrors natural human annotation order and directly informs the AI architecture
- Real-world data quality:** The taxonomy was applied to 476 real operational complaints collected at RAH, exhibiting class imbalance and annotation ambiguity characteristic of real healthcare data

This document serves as the semantic foundation for all other HCAT Insight documentation. The labels, hierarchy, and dependency structure described here are referenced in the Dataset Documentation, AI System Design, Model Experiments, and Benchmark Results documents that follow.

Document prepared for research and academic publication purposes.
Source of truth: HCAT Insight codebase, SQL Server database, and operational complaint dataset at Rassoul Azam Hospital.